

How chips are made

From beach sand to the chip that runs ChatGPT — and why we are building a better one. Click any step to go deeper. Read this before your first day of coding.

CHAPTER 1

It starts with sand

Every chip in the world — your phone, your laptop, NVIDIA's H100 — starts as ordinary beach sand. Not magnetite sand. Regular quartz sand — silicon dioxide (SiO_2). The journey from that sand to a chip with 80 billion transistors is one of the most complex manufacturing processes humans have ever built.

01



Mine silica sand

Quartzite rock or pure silica sand — 99.9% silicon dioxide. Mined from specific high-purity deposits around the world.

The sand used for chips is not random beach sand. It needs to be extremely pure — high-purity silica deposits. The world's best source is actually the Spruce Pine mine in North Carolina, USA. Most of the world's semiconductor-grade quartz comes from one small town. Fun fact: if that mine stopped operating, the global chip industry would halt within months.

02



Smelt into metallurgical silicon

Heat silica with carbon in an electric arc furnace at 2000°C . Carbon strips the oxygen. You get 98% pure silicon metal.

This process is called carbothermic reduction: $\text{SiO}_2 + 2\text{C} \rightarrow \text{Si} + 2\text{CO}_2$. The result is metallurgical-grade silicon — fine for steel but nowhere near pure enough for chips. Chip silicon needs to be

99.9999999% pure — the nine-nines standard. That requires more refining steps.

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Purify to semiconductor grade

Convert to trichlorosilane gas, distil repeatedly, then decompose back to ultra-pure polysilicon. 99.9999999% pure.

The Siemens process converts silicon to SiHCl_3 gas, which is much easier to purify through fractional distillation — the same principle as purifying alcohol. The gas is then decomposed at high temperature back to pure silicon. This step removes impurities down to a few parts per billion. One foreign atom per billion silicon atoms can ruin a transistor.

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Grow a silicon crystal — the Czochralski process

Melt polysilicon at 1414°C . Dip a seed crystal. Slowly pull upward while rotating. A perfect single crystal ingot grows — up to 300mm diameter, 1 metre long.

This is one of the most precise manufacturing processes in existence. The entire ingot must be a single crystal — every silicon atom perfectly aligned. Any defect or grain boundary ruins the electrical properties. The pulling speed, temperature, and rotation rate are all computer-controlled to nanometre precision. The result looks like a shiny silver log — worth around \$50,000.

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Slice into wafers

Use a diamond wire saw to slice the ingot into wafers 0.75mm thick. Polish both sides to atomic flatness. This is the blank canvas for chips.

The wafer surface must be flat to within a few nanometres across its entire 300mm diameter. Any variation and the lithography patterns go out of focus and transistors fail. Polishing uses progressively finer abrasives down to chemical-mechanical planarization — essentially polishing at the atomic level. A single 300mm wafer costs \$100–200 before any chips are made on it. An advanced EUV wafer run costs \$20,000.

SIMPLE ANALOGY

Think of the wafer as a **blank canvas**. Silicon is your painting surface. The transistors you will build on it are the painting. Before you can paint anything, the canvas needs to be perfectly flat — more perfectly flat than anything else humans make.

The transistor — the atom of computing

A transistor is the simplest possible switch. It has one job: be ON or OFF. That is it. 1 or 0. But when you put 80 billion of them on a chip and connect them cleverly — they can run ChatGPT, render games, and navigate spacecraft. Everything in computing comes from this one simple idea.



What a transistor actually is

Three terminals: **Source** (where electrons come from), **Drain** (where they go), and **Gate** (the switch). Apply voltage to the Gate — electrons flow from Source to Drain. Remove voltage — flow stops. That is ON and OFF. That is 1 and 0. That is the entire foundation of computing.

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Doping — give silicon electrical properties

Pure silicon does not conduct well. Add tiny amounts of other atoms — phosphorus (n-type) or boron (p-type) — to create regions that carry current differently.

N-type silicon: add phosphorus atoms (5 electrons) — extra electrons available to carry current. P-type silicon: add boron atoms (3 electrons) — creates "holes" that act like positive charge carriers. A transistor is built by combining n-type and p-type regions cleverly. The boundary between them controls current flow. This is why it is called a **semiconductor** — it conducts sometimes, not always.

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Photolithography — print the pattern

Coat the wafer with light-sensitive material (photoresist). Shine UV light through a mask with the transistor pattern. Exposed areas change chemically. Wash away — pattern remains.

This is exactly like photography — you are exposing a sensitive surface to light through a pattern to create an image. Except the image is 5 nanometres wide (modern chips) and the "camera" uses extreme ultraviolet light with a wavelength of 13.5nm. The EUV machine that does this costs \$150 million and is one of the most complex machines ever built. Only one company in the world makes them: ASML in the Netherlands.

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Etching — carve the structure

Use chemicals or plasma to eat away exposed silicon areas. The photoresist protects the areas you want to keep. Result: 3D structures at nanometre scale.

Modern transistors (FinFETs and Gate-All-Around) are 3D structures — tiny fins of silicon rising from the wafer surface. The gate wraps around the fin to control current from multiple sides. This allows

better control and less leakage at tiny scales. The etching must be precise to within a few atoms. One angstrom off and the transistor does not work.

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Metallisation — connect everything

Deposit thin metal layers (copper, tungsten) to wire all transistors together. Modern chips have 15+ layers of wiring stacked on top of each other.

The wiring is more complex than the transistors themselves. An H100 has 80 billion transistors connected by a metal interconnect network denser than any city road map. The top metal layers are thick for power delivery. Lower layers are ultra-thin for signals. Chemical Mechanical Planarization flattens each layer before the next one is deposited. The total wire length in a modern chip, if unrolled, would stretch for kilometres.

SIMPLE ANALOGY

A transistor is like a **tap controlling water flow**. The Gate is the tap handle. Turn it — water (electrons) flows from Source to Drain. Turn it off — flow stops. Now imagine 80 billion taps on a chip the size of your thumbnail, all turning on and off billions of times per second. That is a modern GPU.

CHAPTER 3

From wafer to chip

One wafer contains hundreds of chips printed side by side. After all the transistors and wiring are made, the wafer gets tested, cut apart, packaged, and tested again. This process takes 3 months and involves over 1,000 individual steps.

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Wafer testing — find the good ones

Automated probes test every chip on the wafer. Mark the dead ones with ink. Yield rate — percentage of working chips — determines profitability.

For a new chip design, yield starts low — sometimes 10-20% of chips work. As the process matures, yield improves to 70-80%. NVIDIA's H100 at 4nm has a yield of around 70% — meaning 30% of every expensive wafer is thrown away. This is a huge cost driver. Your AQQAI chip's yield will start low — improving it is one of the most important engineering challenges.

02



Dicing — cut the wafer into individual chips

Diamond blade or laser cuts between chips along scribe lines. Each individual chip is called a die. Handle with extreme care — they shatter easily.

The H100 die is 814mm² — about the size of a credit card corner. Larger dies have lower yield because defects are more likely to hit the die. This is why chipmakers have moved to chiplets — multiple smaller dies connected together rather than one giant die. NVIDIA's Blackwell uses two dies connected by NVLink. AMD pioneered chiplets; it is now the dominant approach for high-end chips.

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Packaging — put it in a protective case

Bond the die to a substrate. Add heat spreader. Connect memory (HBM) using through-silicon vias. The package protects and connects the chip to the circuit board.

The H100 package includes the GPU die plus 6 stacks of HBM3 memory connected by thousands of tiny bumps called C4 bumps and through-silicon vias (TSVs). The package is larger than the die — it provides mechanical protection and routes the hundreds of power and signal connections. Advanced packaging (CoWoS at TSMC) for H100 costs \$1,600 per chip — just for the packaging step.

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Final test — burn-in and qualification

Run each chip at high temperature for hours to catch early failures. Test at rated speed. Bin chips by maximum clock speed. Ship.

Chips that cannot reach full speed are sold as slower variants — this is why the same die produces different product tiers. An H100 that cannot reach full clock speed becomes an H800 or A100. This binning practice maximises the value extracted from each wafer. The entire fabrication process from wafer start to shipping takes approximately 3 months.

CHAPTER 4

From chip to AI

A GPU is just a chip with a specific architecture designed for parallel math. AI needs parallel math. That alignment — not any magic — is why GPUs run AI. Understanding this is essential for what we are building at AQQA.



Why AI needs parallel math

Running a neural network is essentially **trillions of matrix multiplications**. Multiply a matrix by another matrix. Repeat. A CPU does this one calculation at a time — fast per calculation, but sequential. A GPU does **thousands simultaneously** — each CUDA core handles one small piece. 18,432 CUDA cores in an H100 all working in parallel = AI at scale.

WHAT	CPU (INTEL/AMD)	GPU (NVIDIA H100)
Cores	8–64 powerful cores	18,432 simpler cores
Good at	Sequential tasks, logic	Parallel math, AI
AI training	100x slower	The standard
Power draw	65–125W	700W
Cost	\$200–\$600	\$30,000

SIMPLE ANALOGY

A CPU is like **one brilliant professor** who solves problems one at a time, very fast. A GPU is like **18,000 students** each solving one small part of the same problem simultaneously. For AI — which involves doing the same simple math operation on millions of numbers — 18,000 students wins every time.

Why silicon has a ceiling

Silicon transistors are approaching physical limits. Moore's Law — the doubling of transistors every 2 years — is slowing. At 2nm, transistors are only 10 atoms wide. Quantum effects cause electrons to tunnel through barriers that should stop them. The physics is fighting back.

01



Moore's Law is ending

Transistors cannot shrink forever. At 2nm they are 10 silicon atoms wide. Quantum tunnelling causes electrons to leak even when the transistor is supposed to be OFF.

Gordon Moore predicted in 1965 that transistor count would double every 2 years. It held for 50 years. Now it is slowing — going from 3nm to 2nm takes 4 years, not 2. Each new node costs exponentially

more. The next node after 2nm (1.4nm — called A14) may cost \$30,000 per wafer. At some point the economics break — more transistors cost more to make than the performance gain is worth.

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The power wall

More transistors = more heat. H100 at 700W is near the practical limit of what liquid cooling can handle. You cannot double transistors again without doubling heat — and you cannot cool twice as much heat.

Power density in modern chips has reached the point where some areas of the die are as hot as the surface of the sun — briefly, during switching. Cooling is now the primary constraint on chip performance, not transistor count. This is why every AI data center looks like a refrigerator factory. Microsoft and Google are experimenting with immersion cooling — submerging entire servers in dielectric fluid.

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The cost wall

A new chip fab costs \$20 billion to build. An EUV machine costs \$150 million. Only 3 companies in the world can afford cutting-edge chip manufacturing: TSMC, Samsung, Intel.

The concentration of chip manufacturing in 3 companies — and most cutting-edge work in one (TSMC in Taiwan) — is a geopolitical risk that governments are spending hundreds of billions to address. The US CHIPS Act (\$52B), EU Chips Act (€43B), and India Semiconductor Mission (\$10B) all exist because of this concentration. AQQAI's magnetite approach could be manufactured more simply — potentially without \$20B fabs.

CHAPTER 6

The solution — spintronics

What if instead of moving electrons to store 0 and 1 — we used the spin of electrons? Electron spin is a quantum property — it points either UP or DOWN. That is your 1 and 0. No electron needs to move. Almost no energy needed. No heat generated. This is spintronics — and it is what AQQAI is building with.



What is electron spin?

Every electron acts like a tiny magnet — it has a magnetic moment that points either UP or DOWN. This is a fundamental quantum property called spin. **Spin UP = 1. Spin DOWN = 0.** To flip between states, you apply a magnetic field — almost no energy. Compare this to silicon where you move electrons (charge) — which generates heat every single time.

PROPERTY	SILICON (CHARGE)	SPINTRONICS (SPIN)
How 0/1 stored	Electron present / absent	Spin UP / spin DOWN
Energy to switch	10–50 femtojoules	0.14 femtojoules (100x less)
Heat generated	High — 700W for H100	Near zero
Memory when off	Lost — needs constant power	Permanent — like a magnet
Memory read speed	~100ns (HBM3)	~10ns (10x faster)

SIMPLE ANALOGY

Silicon is like a **light switch that sparks every time you flick it** — the spark is the heat. Spintronics is like a **compass needle** — flipping it from north to south takes almost no energy, generates no heat, and it stays pointing whichever way you left it even when you walk away.

CHAPTER 7

The AQQAI chip

This is what you are building. A chip that uses magnetite — iron oxide from Indian beach sand — as its magnetic memory layer, combined with silicon for fast logic. The result beats the H100 on memory speed and power consumption, while costing a fraction of the price to manufacture.

01 Extract magnetite from Indian beach sand

Fe_3O_4 — iron oxide — naturally concentrated on Kerala, Odisha, Tamil Nadu coastlines. Simple magnetic separation. Near-zero raw material cost.

India has some of the world's richest black sand deposits — naturally concentrated by wave action over millennia. The extraction process is simple — drag powerful magnets through wet sand, magnetite clings, quartz falls away. Industrial processes already do this. Purifying to semiconductor grade (99.999%) is the harder step but uses known chemical processes adapted from silicon purification.

02 Deposit as nanometre-thin film

Use atomic layer deposition (ALD) to grow magnetite in layers just a few atoms thick on a silicon substrate. Precise control of thickness and composition.

ALD works by alternating between two gas-phase precursors — each cycle deposits exactly one atomic layer. For magnetite, iron-containing precursors alternate with oxidising agents. Research in 2025 showed successful Fe_3O_4 ALD below 200°C — compatible with standard CMOS manufacturing. The challenge is maintaining the correct $\text{Fe}^{2+}/\text{Fe}^{3+}$ ratio for good magnetic properties.

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Align magnetic domains — the critical step

Heat in a strong magnetic field (field annealing). All magnetic domains align in one direction. Now the material can store reliable 0s and 1s using spin direction.

This is the unsolved step — and the most important research challenge at AQQAI. Magnetite naturally has random domain orientations. We need all domains pointing the same way for reliable transistor operation. Field annealing gives partial alignment. The December 2025 altermagnetism breakthrough showed that growing thin films with specific crystallographic orientation on sapphire substrates can achieve aligned domains. Adapting this to magnetite on silicon is the key research question.

04



Pattern into magnetic tunnel junctions

Use photolithography to etch the magnetite film into MTJ structures — two magnetic layers separated by a thin insulator. Current flows easily when both layers align (1), blocked when opposite (0).

Magnetic tunnel junctions are already in commercial production — Everspin makes MRAM chips using CoFeB/MgO MTJs. The tunneling magnetoresistance ratio (TMR) — how much current changes between 0 and 1 states — determines reliability. Modern CoFeB MTJs achieve >300% TMR. Magnetite MTJs have been demonstrated in labs but not yet optimised to the same level. Improving TMR ratio is a core research objective.

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Integrate with silicon logic — the AQQAI chip

Silicon transistors handle fast logic. Magnetite MTJ arrays handle memory — 10x faster reads than HBM3, zero power when idle. AI weights burned in permanently. No server needed.

The final architecture: a silicon compute layer (CUDA-compatible cores) with a magnetite memory layer replacing HBM. AI model weights are written once into the magnetite memory — they stay permanently without power, like a magnet. Inference reads weights directly at 10ns latency vs HBM3's 100ns. Power draw drops from 700W to under 50W. Cost drops from \$30,000 to \$5-15 at scale. This is the chip that replaces NVIDIA.

Quick check — can you answer these?

1. What is the raw material silicon chips start from?

Beach sand — silicon dioxide (SiO_2). Purified to 99.9999999% pure silicon, grown into a single crystal, sliced into wafers.

2. What does a transistor do?

It is a switch — ON (1) or OFF (0). Apply voltage to the Gate and electrons flow from Source to Drain. Remove voltage — flow stops.

3. Why does the H100 use 700W?

Every time a silicon transistor switches between 0 and 1, electrons move and generate heat. With 80 billion transistors switching billions of times per second — enormous heat.

4. What is the difference between charge-based and spin-based computing?

Charge-based (silicon): move electrons to store data — generates heat. Spin-based (spintronics): flip electron spin direction — almost no energy, no heat, state persists without power.

5. What is the one unsolved problem in the AQQA chip today?

Atomic-level magnetic domain alignment of magnetite at chip scale and manufacturing volume. Once this is solved, the rest of the architecture follows from existing research.